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Question Attempted	This assignment addresses two parts: Part A proposes a classical Deep Learning solution to a real organisational challenge in grocery retail: Deep Learning for Self-Checkout Theft Detection in Grocery Retail, and Part B reflects on the HEC Paris study trip experience covering subject insights, intercultural learning, and personal development.
Have you used Artificial Intelligence (AI) in any part of this assignment?	Yes
If you have ticked "Yes" above, please briefly outline below which AI tool you have used, and what you have used it for. Please note, you must also reference the use of generative AI correctly within your assessment, in line with the guidance provided in your student handbook. I have used generative AI tools to support the refinement of this assignment. Specifically, AI was used to assist with improving clarity of expression, polishing wording, and checking grammatical accuracy. It also helped in articulating and structuring my own ideas more clearly. The core ideas, analysis, and arguments presented in this assignment are entirely my own and were developed independently based on my understanding of the module content and the interview conducted. No AI-generated content has been used without critical evaluation and adaptation. AI tool used: Claude	

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PART A: Case Study

Seeing What the Till Cannot: Deep Learning for Self-Checkout Theft Detection in Grocery Retail

1. The Business Problem: What the Weight Sensor Cannot See

In 2024/25, retail theft in the UK amounted to £2.2 billion, with the number of shoplifting crimes recorded by the police increasing to its highest level since records began, rising by 20% in a year to over 530,000 (BRC, 2026; ONS, 2025). The problem lies at the heart of self-checkout: independent research concludes that shrink rates at self-checkout lanes are 3.5–4% compared to 0.21% at staffed registers, which is up to sixteen times higher (SAI Group, 2025). Understanding why requires precision about what current safeguards can and cannot detect. Fixed self-check-out terminals have a weight sensor bagging area; scan one item, bag one item, weight should match. This is very effective against quantity fraud; if someone bags two items, and one is scanned, this will alert the system. But it fails to deal with identity theft. A weight sensor only measures the weight of an object, and does not know what it is. It can be defeated entirely with three situations: product substitution, barcode switching, and the loose produce loophole (where the weight of the item is not fixed on a high price, such as premium chocolate being bagged with a weight of 120g and then a scanned 120g banana being pressed into the scanner for the weight against the bar code). With each transaction, the POS log shows a clear and full transaction.

A second fraud risk, and more structurally vulnerable, is that of Scan and Go (Sainsbury's Smartshop and Tesco's Scan as You Shop), rolled out in their more extensive superstore network. Purchasers walk from aisle to aisle with a handheld device to check off the items they've scanned and then show a summary barcode at the exit gate. There is no weight sensor in anywhere in this trip. The exit gate only confirms payment, it doesn't confirm that the content on the trolley is the same as it was in the scan record. The scam is very simple: scan a £0.60 tin, set a £4.50 jar in front of it, present the receipt and walk away. The exposure is over the whole store area as opposed to a specific terminal, so the loss for each trip is significantly greater.

Importantly, both gaps can be closed with available hardware. At Sainsbury's and Tesco, overhead cameras are fitted around self-service checkout and store aisles to be used for post incident review, but not in real time to detect incidents. The lack of infrastructure is not the issue; it's intelligence. This is exactly the type of problem class for which Deep Learning is needed: The data is already there, but it is lacking the means to interpret the data dynamically (HEC Paris, 2026). The weight sensor answers how many, a CNN answers which one.

2. The Proposed Solution: Visual Identity Verification

The proposed solution is a CNN-based computer vision system operating across two contexts, fixed terminals and Scan and Go aisles, asking the same question continuously: does the product the camera identifies match the product the scanner just registered? If it doesn't, an evidentiary alert is sent to a human employee that has a video clip. The machine alarms and the person makes the decision.

Two coordinated CNN models are implemented on over-the-camera terminals at fixed terminals. The first is based on a YOLO (You Only Look Once) architecture where objects are detected as they are manipulated and transferred to the bagging area within a single pass of a neural network without any perceptible lag (Redmon et al., 2016). The second compares the object detected to the retailer's SKU image database. If the visual identity and simultaneous POS stream do not match, a confidence-weighted staff alert is generated. Not instead of the weight sensor, but in parallel: the weight sensor detects quantity fraud, the CNN detects identity fraud. For Scan and Go, the same pipeline is used, with ceiling-mounted cameras in the aisles tracking what is actually being lifted off the shelves and put into trolleys and compared to the device's running transaction log. This follows the deployment shown by Trigo, who was named at the 2026 Retail Technology Awards Europe (Retail Gazette, 2026) for its CNN-based store monitoring platform.

Data is ingested via API into a shared data lake: the data comes from the cameras, the event streams on the POS, the log data of the Scan and Go devices, and the product master database (30,000+ SKUs with seasonal variants in packaging). An ETL layer synchronises all streams by timestamp, resulting in training examples being paired, viz. visual identity vs scan record, for both deployment situations. Model training takes place on cloud GPU infrastructure, with trained weights deployed on edge inference devices at terminals and at in-store edge servers in the aisles, maintaining an latency of less than 200 millisecond. Preprocessing (lighting normalisation, occlusion handling, class-imbalance SMOTE equivalent oversampling) takes about 80% of the project effort as observed in HEC. Train/validate/test=70:15:15, where test is selected from stores not used in training.

3. Risks and Limitations

The biggest risk is the wrongful allegation of an innocent customer. This requires a system that doesn't automatically intervene: All alerts must be investigated by a human staff member who will decide what to do based on the video information. The machine flags, and the human acts. This is not a design preference but an ethical requirement, Floridi (2014) notes, of maintaining "meaningful human agency over consequential decisions delegated to intelligent systems". A similar legal risk is GDPR: Continuous video surveillance of customers is a form of personal data processing, and a policy of data minimisation, limitations of retention and informing customers about use of video is required.

Algorithmic bias is a second ethical risk. When there are historical patterns of discrimination among different groups in the training set, the model may show this discrimination at a machine speed (Mittelstadt et al., 2016). To do the work of mitigation, training should only include objective scan-mismatch events and ongoing bias audits should be performed. Grad-CAM explainability tools which give a visual heat map of the model prediction enable reviewers to ensure that the model is reacting to the product rather than irrelevant parts of the scene (Chattopadhyay et al., 2018). From a technical standpoint model drift is persistent: packaging introduces difference in training distributions and thus, there is a need for continuous training pipelines rather than a single deployment. Energy and infrastructure costs across hundreds of edge devices must also be assessed, the HEC programme highlighted that large-scale AI infrastructure carries material environmental cost that belongs in any responsible business case.

4. Comparison with Alternatives

Human attendants are still the best option, more contextually sensitive, can do discretion and de-escalation. However, the staffing savings that self-checkout was supposed to provide are lost if a single attendant is responsible for six terminals and has to attend to all of them at once. Small-format, low-transaction convenience stores are an excellent place to adopt human oversight.

Classical Machine Learning applied to transaction data, flagging anomalously low spend relative to basket, is interpretable, auditable, and does not need any camera infrastructure. It has the major disadvantage of only being able to read what the scanner captured. It's a symptom of theft, not the act. The weight tolerance rules, which most major retailers already use, are easy to work around, cheap, and transparent: Organised thieves use the rules to their advantage, and the rules are understood by all. In fact, as the HEC programme has established, rule-based systems are unsuccessful just where the issue space is dynamic and combative, and retail theft is. Deep Learning is better suited when camera infrastructure is available or could be cost-justified, product diversity matters, and the shrinkage loss scale is large enough to cover the deployment, maintenance, and risk of false positives. If those conditions do not exist, then simpler isn't a compromise, it's the right thing to do.

5. Business Case

The key stakeholders are the Chief Financial Officer (loss quantification and ROI), the Chief Technology Officer (infrastructure), the Loss Prevention Director (operational deployment), store managers (alert triage) and store associates (front-line alert users). If a 20% cut results in a UK grocery chain with 300 stores losing £60m in annual shrinkage losses can get £12m back per year, that's only a conservative estimate, after similar results from AI monitoring at Walmart (Evinent, 2026). The payback is less than 12 months for a deployment cost of £5 - £8 million for edge compute and model development. Most importantly, the capital cost is less than similar AI projects, since the camera equipment is already in place, the investment is mostly software and edge inference.

The execution risk is also minimized with technology maturity. CNN-based identity verification on existing CCTV infrastructure has been commercialized at the supermarket scale and has been proven by Trigo at the Retail Technology Awards 2026 (Retail Gazette, 2026). The decision as to go/no-go is made at six months depending on the recall level and false positive rates from a pilot across fifteen stores with the aim of minimising capital risk and building the evidence base for chain-wide rollout. This reflects the experimental, iterative mindset that Professor Fraitot explained at HEC as the best managerial attitude towards adopting AI.

For organisational adoption, the system must be positioned as a decision-support tool for staff, not a surveillance tool for customers. The CNN alerts, the staff member judges. This framing matters practically: the 2026 EuroShop conference identified cultural resistance as the primary barrier to AI adoption in retail (EuroShop, 2026). A system that supports staff, not monitors customers, is much more likely to be adhered to, and to produce the results that it promises. The deeper question is accountability: If a machine tags a customer as a likely thief, and a staff member follows through, who is responsible if the customer is innocent? that both Floridi (2014) and Mittelstadt et al. (2016) would ask. The answer needs to be clear on human-readable video evidence and in the system design and governance: the staff member was the guilty party, not an algorithmic score. No responsibility can be passed to the model. Once it is made, the system has entered the realm of autonomous judgment, where current Deep Learning interpretability cannot ethically go.

PART B: Where Data Meets the World: Reflections from HEC Paris

A personal account of learning, culture, and ambition; from the lecture halls of HEC to the corridors of Station F

1. Insights from Subject Experts and Industry Visits

Before joining HEC Paris, I knew AI mostly as a technical field designed by engineers and mostly as a business result. That changed in the 5 days spent in Professor Vincent Fraitot's Data Science & AI for Business programme. So what I was left with, and what I found to be more managerially useful, was a benchmark for when AI is valuable, when it is not, and how people who are deploying it need to behave.

The biggest conceptual change was a simple thought on the first day. Professor Fraitot saw the data as a strategic resource that had to follow a deliberate path from raw input to insight to decision to generate business value. This reframed AI as a management challenge as much as a technical one. Problem definition, data quality, stakeholder alignment and human oversight are often the tricky part of the algorithm. That lesson feeds directly into my dream job of being an AI Product Manager, a position that simply exists to bridge the gap between what AI can do and what a business needs.

The Deep Learning session clarified the conflict I find most intriguing: accuracy vs. interpretability. A CNN can identify a product mismatch with great accuracy but cannot explain it, without the help of tools such as Grad-CAM. Professor Fraitot was to the point: It's not an inconvenience, it's an ethical and a management problem. If the machine makes a consequential decision that the operator and/or the subject cannot question, then it becomes critical to be able to take accountability, which Floridi (2014) calls moral agency. I don't have an answer to the question, but I now recognize that the question must be asked.

The best part of the trip was visiting Station F on Thursday. I wandered through the world's biggest start-up campus, and I found what a lecture room can't offer: the texture of productive ambiguity. The start-ups there were not waiting for certainty before building, they were iterating, failing, and rebuilding, precisely the experimental posture Professor Fraitot described as the right attitude toward AI in business. The Data in Startups session confirmed this: Value is not always in the most complex model, but in the fastest pace from question to experiment to answer. That was a lesson that someone who wanted to pursue a career in AI product management would find very illuminating: The more important question than how much you know, is how fast.

2. Intercultural Learning and Cultural Awareness

I was born and educated in India, then gained my MSc at Warwick, UK, and so came to HEC Paris with two cultural luggage: one from India, one from the UK, and both were slightly and sometimes deeply disoriented by the French academic system.

The striking difference was with India. In Indian academic culture, especially at the postgraduate level, there is a tendency to be deferential to expertise, in the sense that the professor dictates, the student learns, and public criticism, though not completely absent, can be socially dangerous. This is a small part of what Hofstede (2001) would call high power distance, that is, a deeply ingrained attitude that sees hierarchy as legitimate and that regards intellectual authority as being vested in seniority and credential. It's been a different experience at HEC. The sessions were structured and the professor was a leader, but students challenged arguments directly, with no apology, and disagreement was seen as a form of respect and not a social faux pas. That's a real adjustment for someone who has a background in an environment that values that kind of directness in such a way that it's confrontational.

The differences in academic culture between the UK and France were more muted but informative. The study experience at Warwick has taught me a new norm, of active but guarded engagement, critical but courteous engagement, thoughtful engagement, which is not afraid of the social register of challenge. It is important to use Hall's notion of high-context and low-context communication here. The UK is on the low-context end, with content being clearly stated and argument being the primary mode of expression, but an amount of social convention filters that mode. In my experience at HEC, France is a mix of low-context precision (in content) and toleration for the bluntness (in delivery). The outcome is the style accelerating, ideas are “stress-tested” earlier because you don't have to wait for the kind moment to push back. For someone navigating between Indian, British, and now French professional norms, the HEC week made visible how much energy in cross-cultural collaboration is spent on reading the room rather than engaging the argument (Trompenaars and Hampden-Turner, 1997).

This reflection took several turns in the cultural visits. Its sheer architectural ambition, designed specifically to showcase centralised state power in physical terms is a factor that made me think of parallels in both the AI governance discussion and in India's own relationship with centralised institutional authority. In India, the federalism is democratic, while in France it is Napoleonic in its approach to the question of power. At organizational level, large-scale machine learning platforms are a technology that brings all analytical power and ask the same question. The evening in Montmartre was a counterpoint of an irreverent, public and democratic, definitely French, nature of creativity and intellectual life. I was reminded of the streets of Kolkata or Varanasi ghats, places where the cultural life isn't curated for consumption in public space, as I climb towards Sacré-Cœur. I didn't expect the parallel and for a split second Paris seemed less foreign than I thought.

3. Personal Reflections and Observations

I came to Warwick from India with a clear ambition to transition into AI product management, but also with an awareness that the distance between knowing what you want and knowing how to get there can feel large. The HEC Paris sessions, the trip to Station F, discussions with student ambassadors, and the course dinner on Thursday night have helped me narrow down that gap.

There was one particular incident during our visit to Station F that I would like to mention here. Eliot Sberro, a co-founder of Univia.AI, a startup created in the incubator of HEC Paris and focused on applying AI solutions to solve actual business problems, showed us his project and then completely turned the tables on us, asking us to come up with and pitch our ideas. That particular experience was not very big, but it was quite eye-opening. There is a difference between learning about AI products and being challenged to come up with a solution on your own and present it to the founder of a company who is looking for innovations.

Interactions with student ambassadors from HEC during the programme gave me a new perspective on what international professional education entails. The fact that these students were comfortable switching between technical and managerial jargon, talking about boardroom dynamics as well as data dynamics, was evidence that dual fluency not only achievable but expected. Being a student who has had experience with two different educational cultures, one at an undergraduate level in India, and another at a postgraduate level in Britain, I could see in them the very traits that I was trying to develop at Warwick.

The course dinner, after the Montmartre visit, provided an unexpectedly rich learning moment. Conversation ranged from the ethics of algorithmic decision-making to career anxieties shared openly and genuinely funny disagreements about what constituted adequate cultural immersion. It was a reminder that durable professional networks are built in unstructured moments where people are being themselves, a skill as worth developing as any technical capability. Perhaps the most unexpected discovery of the whole trip was this: that the most useful things I learned did not happen during the scheduled sessions, but in the conversations around them, at lunch tables, on buses, and over dinner, which suggests that the design of international learning experiences matters as much as their content.

If I were to name a single commitment I am taking from Paris: AI is not a solution, it is a capability. It creates value only when deployed against a precisely defined problem, with good data, honest acknowledgement of its limits, and humans who remain accountable for its outputs. Professor Fraitot noted that Data Science is still an emerging field and managers should approach it experimentally. I would extend that: the experiment is not only technical, it is organisational, ethical, and personal. Paris gave me a clearer sense of what my version of that experiment looks like.

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